

Advaith Prabhu

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RESEARCH & PROJECTS

ML-Filtered EMA Momentum Strategy

Remote

Author - Submitted to American Journal of Capstone Research

Dec 2025-Present

- Designed and backtested an MLP confidence filter applied to EMA(20/50) crossover signals across 12 S&P 500 equities (2010–2025), reducing mean maximum drawdown by 5.81pp while preserving signal integrity.
- Engineered 7 lag adjusted features (RSI, ATR, volume z-score, multi-horizon returns, volatility) in Python to eliminate look-ahead bias; implemented 60/20/20 chronological train-val-test split across 15 years of daily price data.
- Quantified a calibration-consistent accuracy regime (~52% mean test accuracy) to argue that the filter functions as a risk-suppression tool rather than a return enhancement tool; an non-obvious distinction framed via Sharpe decomposition.

MCMC Approach on Collatz Map

Remote

Independent Research Project

April 2025-Present

- Constructed a Markov Chain over residue classes derived from the accelerated Collatz Map, proving the transition matrix is a 0-1 permutation matrix with analytically stationary distribution.
- Implemented a Metropolis-Hastings sampling on the Collatz Map in Python (numpy/scipy), pairing each algorithm with a formal proof of correctness.
- Developed a 5 module codebase with full code-proof pairing, targeting undergraduate-level mathematical rigour in an independent setting.

World Studies Extended Essay - Monte Carlo Options Pricing

Remote

IB Extended Essay

2025-2027

- Modelled European options prices using Monte Carlo simulations of geometric Brownian Motion paths, benchmarking simulated prices against Black-Scholes analytical value, with respect to convergence rate and path count.
- Implemented a simulation engine in Python, demonstrating the prices estimate converge to within 0.5% of the Black-Scholes values at $N > 10,000$, providing empirical grounding to the theoretical law of large numbers result

COMPETITIONS & AWARDS

Awards

Singapore

FedEx JA International Trade Challenge 2025

Top 5

International Business Olympiad

Higher Distinction

EDUCATION

Global Indian International School

Singapore

IB Diploma - Economics HL, Computer Science HL, Mathematics AA HL

May 2027

- Extended Essay on How Monte Carlo simulations affect European options pricing

SKILLS & INTERESTS

Skills: Languages: Python (pandas, numpy, scipy, scikit-learn, matplotlib), LaTeX, SQL (basic); Methods: MCMC, Monte Carlo simulation, MLP classification, time-series backtesting, Black-Scholes pricing; Tools: yfinance, Git/Github. AI: Claude Code (skills, MCP, RAG, Obsidian)

Interests: Watching movies, reading classical fiction books